1. Probability Basics

Theory

Probability kisi event ke hone ki possibility ko measure karti hai. Iska value hamesha **0 se 1** ke beech hota hai.

- **0** → Event impossible hai
- **1** → Event sure hai
- **0.5** → 50-50 chance

Sample Space (S): Experiment ke saare possible outcomes ka set.

Example: Fair die → S = {1, 2, 3, 4, 5, 6}

Event (E): Sample Space ka koi subset.

Example: Even number aana → E = {2, 4, 6}

Joint Probability: Do ya do se zyada events ke **ek saath** hone ki probability. Ise P(A∩B)P(A \cap B)P(A∩B) se denote karte hain.

Probability Rules (Axioms)

1. Probability kabhi negative nahi hoti → P(E)≥0P(E) \geq 0P(E)≥0
2. Sample Space ki probability 1 hoti hai → P(S)=1P(S) = 1P(S)=1
3. Agar events mutually exclusive hain, to unki union ki probability unki individual probabilities ka sum hoti hai.

Formulas (Ek-Ek karke)

1. Classical Probability

P(E)=Number of favourable outcomesTotal number of outcomesP(E) = \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}}P(E)=Total number of outcomesNumber of favourable outcomes

2. Addition Rule (Union)

General case (jab events overlap kar sakte hain):

P(A∪B)=P(A)+P(B)−P(A∩B)P(A \cup B) = P(A) + P(B) - P(A \cap B)P(A∪B)=P(A)+P(B)−P(A∩B)

Kyunki P(A∩B)P(A \cap B)P(A∩B) do baar count ho jata hai, isliye subtract karte hain.

Special Case – Mutually Exclusive Events (overlap nahi hota):

P(A∪B)=P(A)+P(B)P(A \cup B) = P(A) + P(B)P(A∪B)=P(A)+P(B)

(kyunki P(A∩B)=0P(A \cap B) = 0P(A∩B)=0)

3. Multiplication Rule (Intersection / Joint Probability)

General case:

P(A∩B)=P(A)×P(B∣A)P(A \cap B) = P(A) \times P(B|A)P(A∩B)=P(A)×P(B∣A)

ya

P(A∩B)=P(B)×P(A∣B)P(A \cap B) = P(B) \times P(A|B)P(A∩B)=P(B)×P(A∣B)

Special Case – Independent Events:

P(A∩B)=P(A)×P(B)P(A \cap B) = P(A) \times P(B)P(A∩B)=P(A)×P(B)

4. Complement Rule

P(Ac)=1−P(A)P(A^c) = 1 - P(A)P(Ac)=1−P(A)

(A nahi hone ki probability)

Mutually Exclusive vs Independent Events

Feature	Mutually Exclusive	Independent
Meaning	Ek saath nahi ho sakte	Ek hone se doosre pe asar nahi padta
Joint Probability	P(A∩B)=0P(A \cap B) = 0P(A∩B)=0	P(A∩B)=P(A)×P(B)P(A \cap B) = P(A) \times P(B)P(A∩B)=P(A)×P(B)
Example	Ek die pe 2 aur 5 ek saath nahi aa sakte	Do alag coins ka toss

Important: Mutually exclusive events independent **nahi** hote (jab tak probability zero na ho).

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2. Conditional Probability

Theory

Conditional Probability batati hai ki **agar ek event pehle se ho chuka hai**, to doosre event ki probability kitni hai.

Example: Agar pehli ball red nikal gayi, to doosri ball red hone ki probability kya hogi?

Yeh concept Data Science me bohot use hota hai (customer behavior, disease prediction, spam filtering etc.)

Formulas (Ek-Ek karke)

1. Conditional Probability

P(A∣B)=P(A∩B)P(B)(P(B)>0)P(A|B) = \frac{P(A \cap B)}{P(B)} \quad (P(B) > 0)P(A∣B)=P(B)P(A∩B)(P(B)>0)

Matlab: A aur B dono hone ki probability ko sirf B hone ki probability se divide karte hain.

2. Chain Rule (Multiplication Rule ka extended form)

Do events:

P(A∩B)=P(A)×P(B∣A)P(A \cap B) = P(A) \times P(B|A)P(A∩B)=P(A)×P(B∣A)

Teen events:

P(A∩B∩C)=P(A)×P(B∣A)×P(C∣A∩B)P(A \cap B \cap C) = P(A) \times P(B|A) \times P(C|A \cap B)P(A∩B∩C)=P(A)×P(B∣A)×P(C∣A∩B)

3. Independence Check

A aur B independent hain agar:

P(A∣B)=P(A)P(A|B) = P(A)P(A∣B)=P(A)

ya

P(B∣A)=P(B)P(B|A) = P(B)P(B∣A)=P(B)

4. Law of Total Probability

Agar sample space ko mutually exclusive + exhaustive parts B1,B2,…,BnB_1, B_2, \dots, B_nB1,B2,…,Bn me baanta ho, to:

P(A)=P(A∣B1)P(B1)+P(A∣B2)P(B2)+⋯+P(A∣Bn)P(Bn)P(A) = P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + \dots + P(A|B_n)P(B_n)P(A)=P(A∣B1)P(B1)+P(A∣B2)P(B2)+⋯+P(A∣Bn)P(Bn)

Short form:

P(A)=∑P(A∣Bi)⋅P(Bi)P(A) = \sum P(A|B_i) \cdot P(B_i)P(A)=∑P(A∣Bi)⋅P(Bi)

Yeh formula Bayes' Theorem ka base hai.

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3. Bayes' Theorem ⭐

Theory

Bayes' Theorem hume **prior knowledge** ko **naye evidence** ke saath update karke **posterior probability** nikalne me help karta hai.

Simple language me:

"Pehle se jo chance thi, usme naya information aane ke baad chance kaise change hoti hai?"

Data Science me iska sabse bada use:

- Naive Bayes Classifier
- Medical diagnosis
- Spam detection
- A/B testing

Formula

Basic Form:

$P(A|B)=P(B|A)\cdot P(A)P(B)P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}P(A|B)=P(B)P(B|A)\cdot P(A)$

Jahan:

- $P(A)P(A)P(A)$ → Prior probability (pehle se knowledge)
- $P(B|A)P(B|A)P(B|A)$ → Likelihood (A hone par B ki probability)
- $P(B)P(B)P(B)$ → Evidence (total probability of B)
- $P(A|B)P(A|B)P(A|B)$ → Posterior probability (naya updated chance)

Extended Form (jab kai events ho):

$P(Ai|B)=P(B|Ai)\cdot P(Ai)\sum jP(B|Aj)\cdot P(Aj)P(A_i|B) = \frac{P(B|A_i) \cdot P(A_i)}{\sum_j P(B|A_j) \cdot P(A_j)}P(Ai|B)=\sum jP(B|Aj)\cdot P(Aj)P(B|Ai)\cdot P(Ai)$

Practice Questions + Step-by-Step Solutions

Question 1 (Addition + Joint Probability)

Ek class me 50 students hain. 30 students Python jaante hain, 25 students SQL jaante hain, aur 15 students dono jaante hain. Ek student randomly select kiya. Uske Python **ya** SQL jaanne ki probability kya hai?

Solution:

$P(P \cup S) = P(P) + P(S) - P(P \cap S) = \frac{30}{50} + \frac{25}{50} - \frac{15}{50} = \frac{40}{50} = 0.8$$P(P \cup S) = P(P) + P(S) - P(P \cap S) = 5030 + 5025 - 5015 = 5040 = 0.8$

Question 2 (Complement + Multiplication)

Ek bag me 8 red aur 5 blue balls hain. 2 balls bina replacement ke nikaali gayi. Dono balls blue **nahi** hone ki probability kya hai?

Solution:

Pehle dono blue hone ki probability nikaalte hain:

$P(\text{dono blue}) = \frac{5}{13} \times \frac{4}{12} = \frac{20}{156} = \frac{5}{39}$$P(\text{dono blue}) = \frac{1}{3} \times \frac{1}{4} = \frac{1}{20} = 0.05$

Ab complement:

$P(\text{dono blue nahi}) = 1 - \frac{5}{39} = \frac{34}{39}$$P(\text{dono blue nahi}) = 1 - 395 = 3934$

Question 3 (Conditional Probability)

Ek company me 60% employees male hain. Male me se 40% aur female me se 25% Python jaante hain. Ek random employee Python jaanta hai. Uske male hone ki probability?

(Pehle Law of Total Probability use karo, phir Conditional)

Solution:

Let M = Male, F = Female, Py = Python jaanta hai

$P(Py) = P(Py|M)P(M) + P(Py|F)P(F) = (0.4)(0.6) + (0.25)(0.4) = 0.24 + 0.10 = 0.34$$P(Py) = P(Py|M)P(M) + P(Py|F)P(F) = (0.4)(0.6) + (0.25)(0.4) = 0.24 + 0.10 = 0.34$

Ab Conditional:

$P(M|Py) = \frac{P(Py|M)P(M)}{P(Py)} = \frac{0.24}{0.34} \approx 0.7059$$P(M|Py) = \frac{P(Py|M)P(M)}{P(Py)} = 0.340.24 \approx 0.7059$

Question 4 (Bayes' Theorem)

Ek diagnostic test ki accuracy:

- Disease hone par positive aane ki probability = 95%
- Disease na hone par positive aane ki probability = 8%
- Population me disease hone ki probability = 2%

Test positive aaya. Actual me disease hone ki probability kya hai?

Solution:

Let D = Disease, T+ = Test Positive

P(D∣T+)=P(T+∣D)⋅P(D)P(T+)P(D|T+) = \frac{P(T+|D) \cdot P(D)}{P(T+)}P(D∣T+)=P(T+)P(T+∣D)⋅P(D)

Pehle P(T+)P(T+)P(T+) nikaalte hain (Law of Total Probability):

P(T+)=(0.95)(0.02)+(0.08)(0.98)=0.019+0.0784=0.0974P(T+) = (0.95)(0.02) + (0.08)(0.98) = 0.019 + 0.0784 = 0.0974P(T+)=(0.95)(0.02)+(0.08)(0.98)=0.019+0.0784=0.0974

Ab Bayes:

P(D∣T+)=0.95×0.020.0974=0.0190.0974≈0.1951(19.51%)P(D|T+) = \frac{0.95 \times 0.02}{0.0974} = \frac{0.019}{0.0974} \approx 0.1951 \quad (19.51\%)P(D∣T+)=0.09740.95×0.02=0.09740.019≈0.1951(19.51%)

Question 5 (Independent Events + Joint)

Do fair dice throw kiye. Pehle die pe 6 aaye **aur** doosre die pe even number aaye ki probability?

Solution:

Independent events hain:

P(1st = 6)=16,P(2nd even)=36=12P(\text{1st = 6}) = \frac{1}{6}, \quad P(\text{2nd even}) = \frac{3}{6} = \frac{1}{2}P(1st = 6)=61,P(2nd even)=63=21
P(dono)=16×12=112P(\text{dono}) = \frac{1}{6} \times \frac{1}{2} = \frac{1}{12}P(dono)=61×21=121

Summary (Students ko yaad dilane ke liye)

- Joint Probability → P(A∩B)P(A \cap B)P(A∩B)
- Addition → Union ke liye
- Multiplication → Intersection ke liye

- Conditional → P(A∣B)=P(A∩B)P(B)P(A|B) = \frac{P(A \cap B)}{P(B)}P(A∣B)=P(B)P(A∩B)
- Bayes → Prior ko Posterior me badalta hai

Aapko in questions ke alawa **aur questions**, **visual explanation**, ya **Python code** chahiye to batao.

Explore Markov Chains theoryStudy Monte Carlo simulations